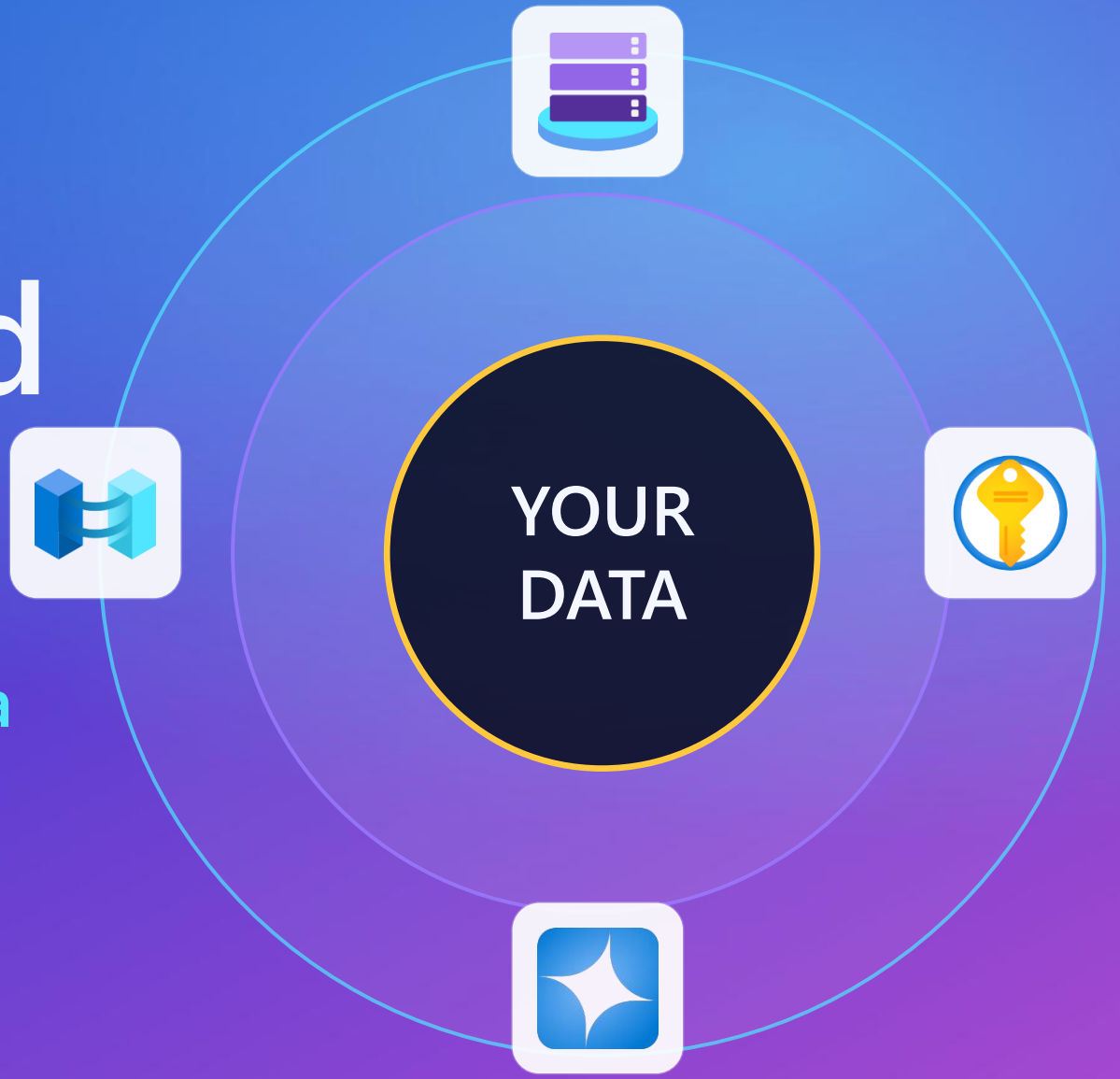
DEVELOPER SUMMIT 2026

Sovereign Cloud & AI

Bringing Cloud and AI to Your Data

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ABOUT ME



Kardel Rüveyda Çetin

Digital Cloud Solution Architect

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Former Microsoft MVP
Community Speaker

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EDUCATION

2013–2018
Yildiz Technical University
Mathematical Engineering
Bahcelor

2019–2020
Yildiz Technical University
Computer Engineering / IT
Master Degree (Non-Thesis)

2023–2025
Yildiz Technical University
Mathematical Engineering
Master Degree (Thesis)

EXPERIENCE

2016–17
Mavi · Overtech
Internships

2018
Iconec Wave
Jr. Frontend Dev

2018–21
Kariyer.net
Software Specialist

2021–25
Dogus Tecnology
SW Spec → Dev Lead

2026–
Microsoft
Digital CSA

Whose data is it, anyway?

Your data is your data



Where does it live?



Who can touch it?



What does the AI see?

AGENDA

Today: from why to Foundry Local on Azure Local

01		**Why** Data, AI and the rules that bind them	5 min
02		**Sovereign Cloud** Microsoft's three pillars	2 min
03		**Azure Local** Hardware, scale, services, connected or disconnected	9 min
04		**Foundry Local** Small models on a device	2 min
05		**Foundry Local on Azure Local** Architecture, deployment, security, disconnected, agents	12 min
06		**Lab: Leak Hunt** Hunt, fix and prove it on this laptop	7 min

About 40 minutes. The lab runs on this laptop; the code walkthrough comes after the talk.

01

Why

Data has gravity. AI wants it. Rules constrain it.

WHY

The sovereignty paradox

AI wants…

More context, more data

GPUs and fast iteration

Managed services and new models

vs

Your data can't…

Leave the country or the datacenter

Be seen by third parties

Be copied freely

Flip the model: don't move data to AI. Move AI to the data.

WHY

Data gravity

> DM
>
> *"As Data accumulates (builds mass) there is a greater likelihood that additional Services and Applications will be attracted to this data."*
>
> Dave McCrory, Data Gravity in the Clouds (2010)



The bigger the data, the harder it is to move.

WHY

Not every workload can go to the public cloud



Latency sensitive

Manufacturing execution, industrial QA and financial infrastructure need near real-time decisions.



AI at the source

Camera feeds, sensors and leak detection must be processed where the data is born.



Regulation and sovereignty

Highly regulated sectors require data and compute to stay under local control.



Business continuity

Production lines, stores and clinics must keep running through network outages.

Cloud-first is not cloud-only.

WHY

Three dimensions of digital sovereignty

Data controls

- Customer-managed keys
- Confidential computing
- Data residency policy

Operational controls

- Customer Lockbox, JIT
- RBAC and PIM
- Audit and transparency

Technological independence

- Azure Arc, open standards
- Hybrid and disconnected
- Portable Kubernetes

HOW

Sovereignty is a spectrum

Less control, less ops → More control, more ops





Public Cloud

Azure regions, RBAC, Policy

Control

Ops effort



Sovereign Public

Residency, CMK/HSM, Lockbox, Confidential VMs

Control

Ops effort



Sovereign Private

Azure Local + Arc in your datacenter

Control

Ops effort



Disconnected

Air-gapped, no cloud dependency

Control

Ops effort

02

Sovereign Cloud

Microsoft's answer: three pillars and the controls behind them

WHY

Microsoft Sovereign Cloud: three pillars



Sovereign Public Cloud

Microsoft-operated Azure regions with defined data boundaries

Key control: customer-managed and external keys

Confidential computing and the Sovereign Landing Zone



Sovereign Private Cloud

Azure Local is the foundation

Connected or disconnected operations

Microsoft 365 Local, Foundry Local, GitHub Enterprise Local



National Partner Clouds

Operated by national partners, under national jurisdiction

Bleu in France: Orange and Capgemini

Delos Cloud in Germany: an SAP subsidiary

One consistent infrastructure management and development platform across all three

03

Azure Local

Azure in your datacenter, at the edge, or fully disconnected

WHY

Sovereign Private Cloud: what is inside



AI suite

Foundry Local
on Azure Local (preview)



Productivity suite

Microsoft 365 Local
GitHub Enterprise Local (preview)



Data suite

SQL Server and
more Azure data services



Your own apps

General purpose VMs
and AKS

PRIVATE CLOUD INFRASTRUCTURE: AZURE LOCAL



Runs with or without cloud connectivity

Connected, or disconnected operations



Flexible deployment and management

Premier, integrated or validated hardware



Select your own scale

One machine up to multi-rack



Resilient by design

High availability and recovery options



Control where your data lives

On your own hardware, on premises



Control who accesses your data

RBAC and a local control plane

AZURE LOCAL

Azure Local: Azure in your datacenter

Cloud-consistent

Same Azure portal, CLI and ARM templates

Runs anywhere

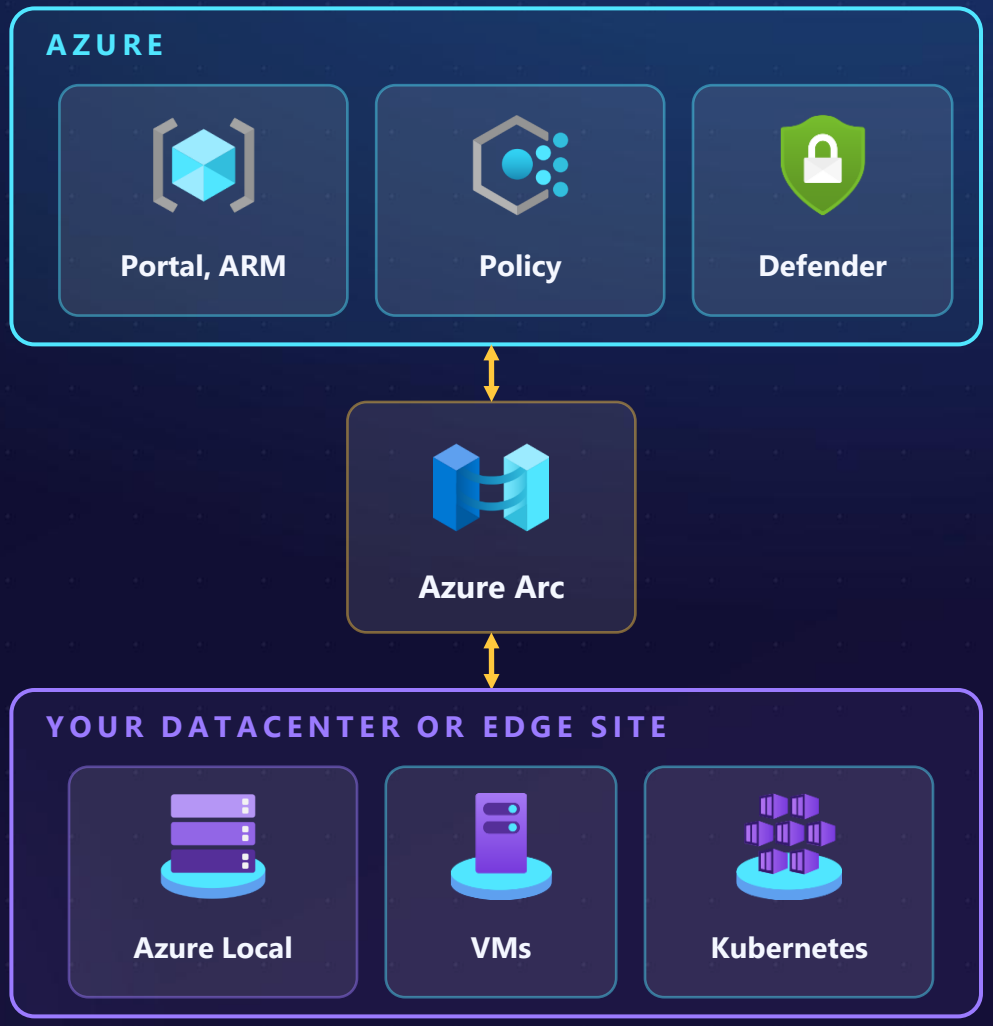
Datacenter, edge or fully disconnected

One control plane

Azure Arc extends Policy, Defender, Monitor

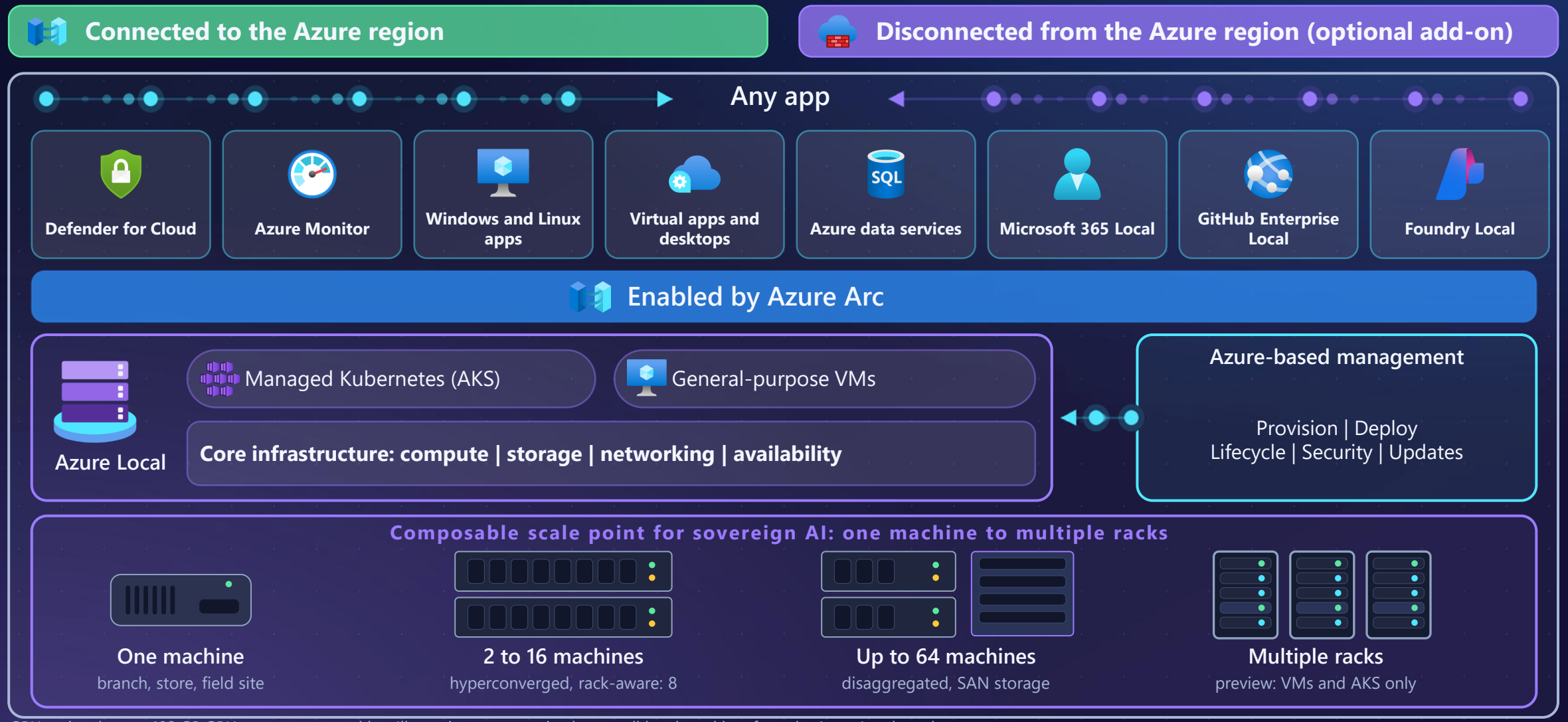
Broad workloads

VMs, Kubernetes, desktops, data services, AI



AZURE LOCAL

Azure Local: AI-ready infrastructure, anywhere



GPU optional: up to 192 GB GPU memory per machine. Illustrations are generic; choose validated machines from the Azure Local catalog.

AZURE LOCAL

Choose hardware from your preferred vendor



Dell Technologies

AX nodes, APEX Cloud Platform



Lenovo

ThinkAgile MX



Hewlett Packard Enterprise

ProLiant DL380 and DL145 Gen11



DataON

Integrated systems



Cisco

Azure Local catalog partner



Supermicro

Azure Local catalog partner



Fujitsu

Azure Local catalog partner

1. Engage directly with your preferred vendor
2. Customize hardware specs, storage and networking
3. Buy validated machines from one node to many



Browse validated solutions: aka.ms/AzureLocalCatalog

Illustrations are generic, not vendor products. Named models are examples from Microsoft Learn; check the catalog for current validated hardware.

AZURE LOCAL

How an Azure Local deployment works



1 Choose hardware

Validated machines from the Azure Local catalog



2 Prepare and install

Prepare Active Directory, download the OS from the portal, install it on each machine



3 Register with Azure Arc

Each machine is registered, with or without the Arc gateway



4 Deploy

From the Azure portal or an ARM template. An Arc resource bridge is deployed automatically



Monitor with Azure Monitor



Update with Azure Update Manager



Defender + Policy

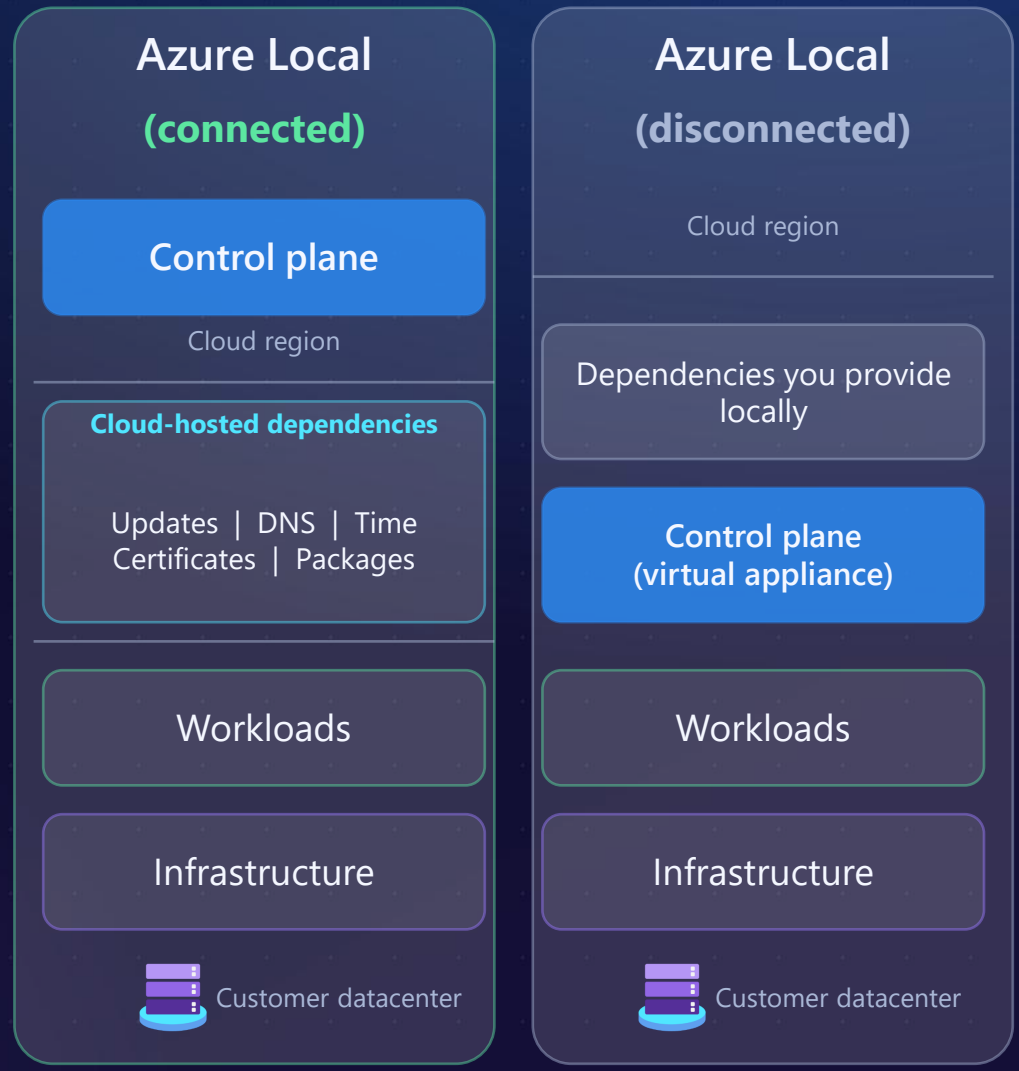
Machine provisioning without installing the OS by hand is in preview.

AZURE LOCAL

Connected vs fully disconnected

Operating fully disconnected is different

- Dependencies you take for granted when connected move to you: network, identity, PKI
- Control plane: Microsoft-operated in Azure vs a local one that your team operates
- A subset of Azure services: portal, ARM, RBAC, VMs, AKS, Registry, Key Vault, Policy
- More hardware: Premier Solutions and a dedicated management cluster
- Eligibility: eligible agreement, support plan, a valid business need
- Your team deploys, monitors and collects logs locally

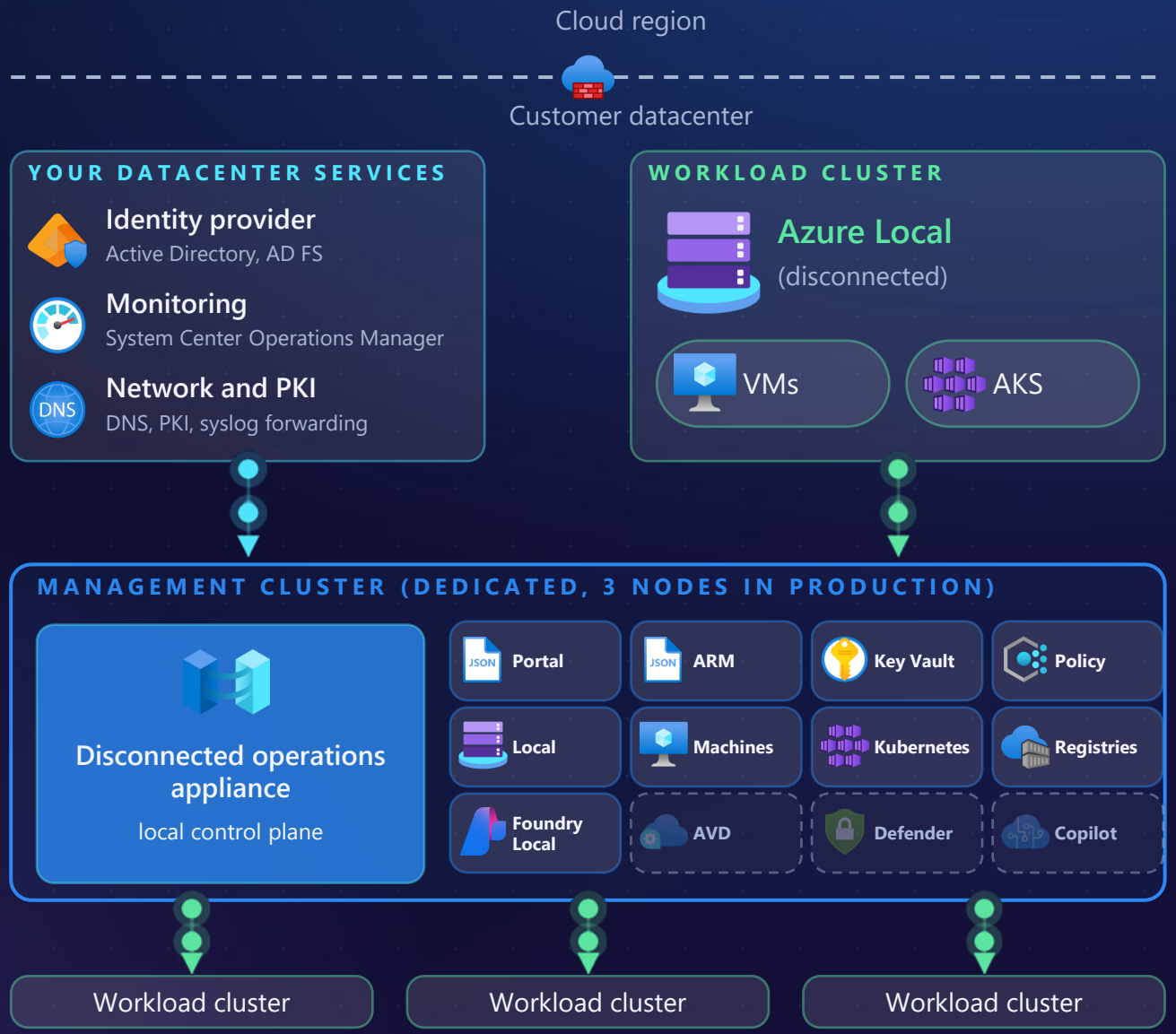


AZURE LOCAL

Disconnected operations: how it is built

- Customer-managed control plane and infrastructure for air-gapped and sovereign scenarios
- Integrates with your identity, monitoring and PKI systems
- Familiar Azure portal, Azure CLI and ARM templates
- Cloud-consistent operating model for your users
- Production needs a dedicated three-node management cluster
- Workloads run on separate Azure Local clusters

A subset of Azure services is served locally. Dashed tiles are not in the documented disconnected supported-services list.



AZURE LOCAL

The AI stack on Azure Local

AZURE LOCAL AI STACK

AI SERVICES

Agentic Retrieval in Foundry Local

Agentic RAG over your documents, built-in MCP server and chat UI, NFS sources

PREVIEW Formerly Edge RAG

Your own apps and agents

Call the OpenAI-compatible endpoint; API key, Entra ID or service account token

PREVIEW Foundry Local on Azure Local

Video Indexer enabled by Arc

Live and recorded video and audio analysis; video data stays on premises

GATED Connected mode only

FOUNDRY LOCAL MODEL CATALOG

PREVIEW

Phi	Mistral	gpt-oss	Llama	DeepSeek
Microsoft	Mistral AI	OpenAI	Meta	DeepSeek

Solid: named as examples on Learn. Dashed: not named there, check the catalog.

Task-specific, fine-tuned or your own models: bring your own model (preview)

FOUNDRY LOCAL MODEL SERVER

ONNX runtime	CPU or GPU	**vLLM runtime**	GPU
Model and ModelDeployment	custom resources	**Evaluation jobs**	scores stored on the cluster

AI INFRASTRUCTURE

NVIDIA GPUs

T4, A2, A16, L4, L40, L40S, RTX Pro 6000; up to 192 GB GPU memory per machine

Leading OEM partners

Dell Technologies, HPE, Lenovo, DataON and more

Small to large footprints

One machine to multiple racks (multi-rack is preview)

Connected and fully disconnected operations (preview)

04

Foundry Local

Small models, close to the data

FOUNDRY LOCAL

Microsoft Foundry spans cloud to edge



CLOUD

Microsoft Foundry

Frontier models and agents

Fine-tuning hub

Breadth and top reasoning



DEVICE

Foundry Local

SLMs on laptops and desktops

SDK, works offline

No Azure subscription needed



ON PREMISES

Foundry Local on Azure Local

Models and inference on your cluster

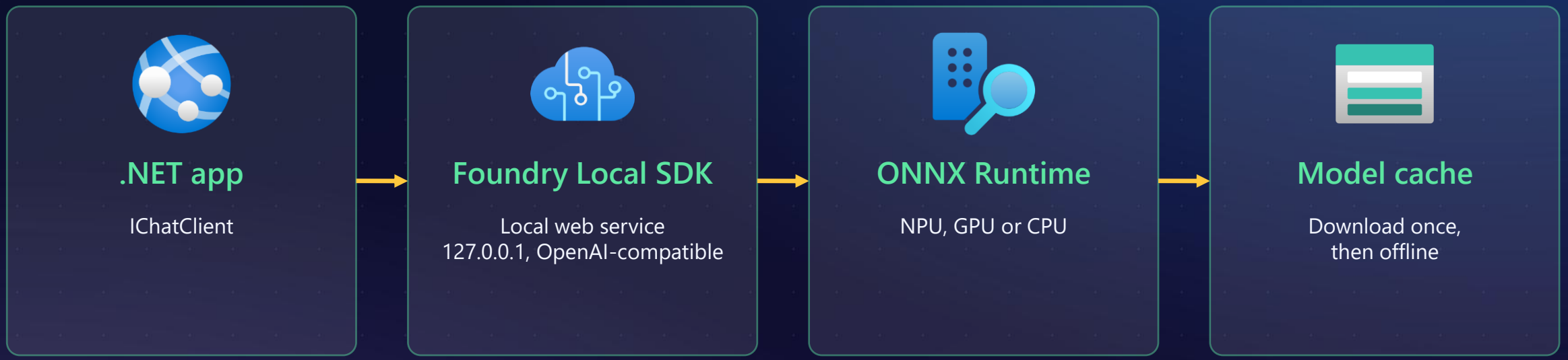
Edge, hybrid or disconnected

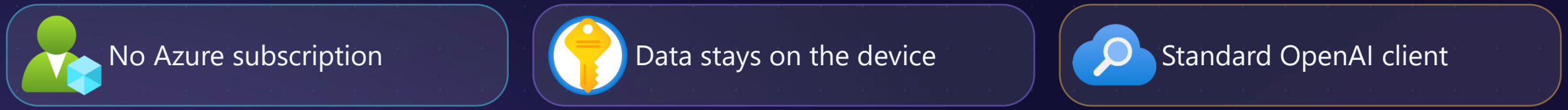
Arc extension (preview)

One OpenAI-compatible API shape across all three. Pick by where the data may live.

FOUNDRY LOCAL

Foundry Local on a device: how it works





While loading, the SDK may open HTTPS connections to Microsoft endpoints (execution provider and model checks). After the first download, models run offline from the local cache.

05

Foundry Local on Azure Local

FOUNDRY LOCAL

Foundry Local on Azure Local: what is in the box

PREVIEW

Models

Catalog models or your own

- Foundry catalog or bring your own model
- ONNX Runtime or vLLM
- One OpenAI-compatible API

PREVIEW

Knowledge

Agentic Retrieval in Foundry Local

- Ingest your documents
- Embeddings and vector search
- Chat with citations

PREVIEW

Tools

MCP for actions

- Built-in MCP server
- Connect your own MCP servers
- The agent acts, not only answers

Enabled by Azure Arc: installed as an extension, managed with Azure tools

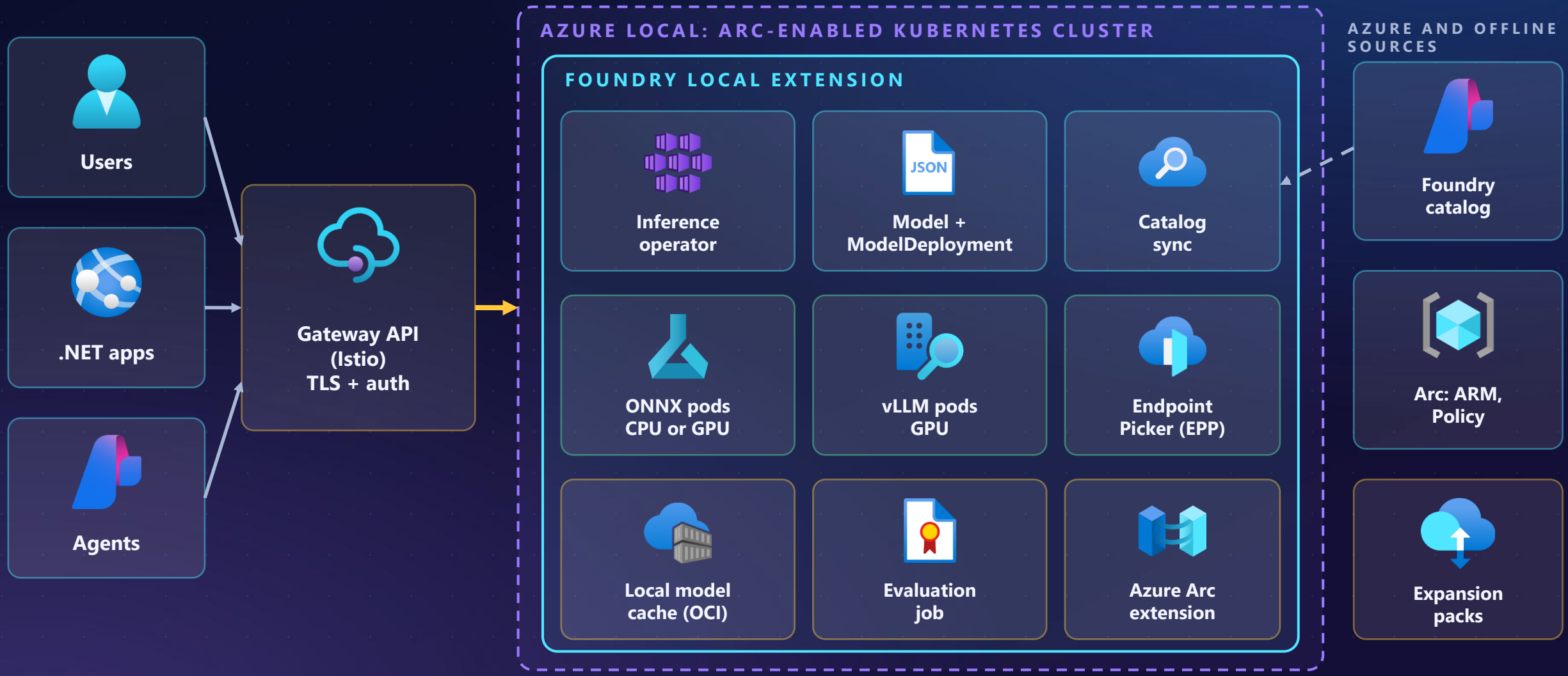
Azure Local: CPU or GPU nodes, in your datacenter or at the edge, connected or disconnected

The model answers. Knowledge brings your data. Tools let it act.

Status per Microsoft Learn, October 2026: Foundry Local on Azure Local and Agentic Retrieval are in preview.

PRODUCTION

Architecture: Foundry Local on Azure Local



PRODUCTION

From zero to first answer: five stages

1 Prepare



Arc-enabled cluster, capacity, GPU drivers, Gateway API, Istio, certificates

FIRST EVIDENCE

```
kubectl get gatewayclass
istio
shows Accepted
```

2 Install



Foundry Local extension from the portal, CLI or Helm, plus the Entra app

FIRST EVIDENCE

```
Extension healthy,
operator pods running
```

3 Choose



A catalog model or your own, sized for your hardware

FIRST EVIDENCE

```
kubectl get configmap
foundry-local-catalog -n
foundry-local-operator
```

4 Deploy



Cache the model, schedule the pod, load the model

FIRST EVIDENCE

```
kubectl get
modeldeployment -n
foundry-local-operator
reaches Running
```

5 Infer



Call the endpoint with the right credential

FIRST EVIDENCE

```
OpenAI-compatible call
returns an answer
```

WHEN SOMETHING BREAKS, NAME THE STAGE FIRST

Cannot install

stage 1 or 2: prerequisites vs extension

Model never ready

stage 4: download, scheduling or load

Request rejected

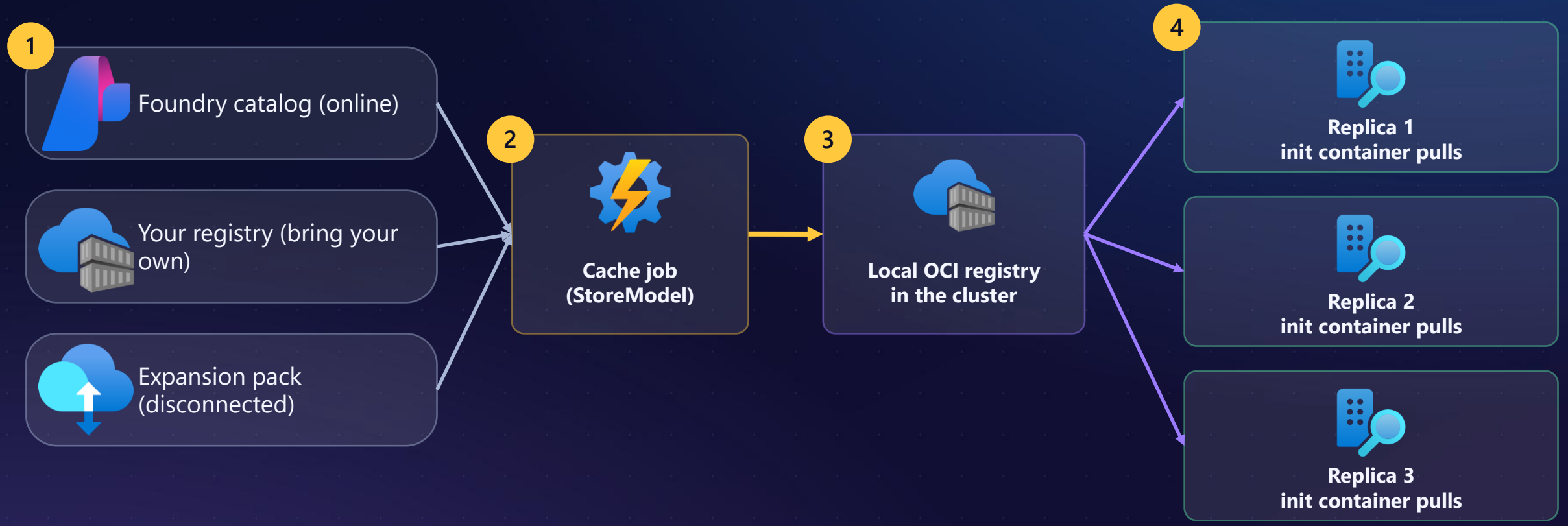
stage 5: endpoint, credential or permission

Broke after an update

compare versions and configuration

PRODUCTION

Model delivery: pull once, serve from inside



Cached once. New replicas pull inside the cluster: fast scale out, no internet needed.

PRODUCTION

Pick the runtime: ONNX Runtime or vLLM

ONNX Runtime

CPU or GPU. The only CPU option

Compact models: Phi-4, Qwen 2.5

Lowest overhead, edge friendly

ONNX format, one model per pod

vLLM

GPU only (CUDA)

High throughput under concurrency

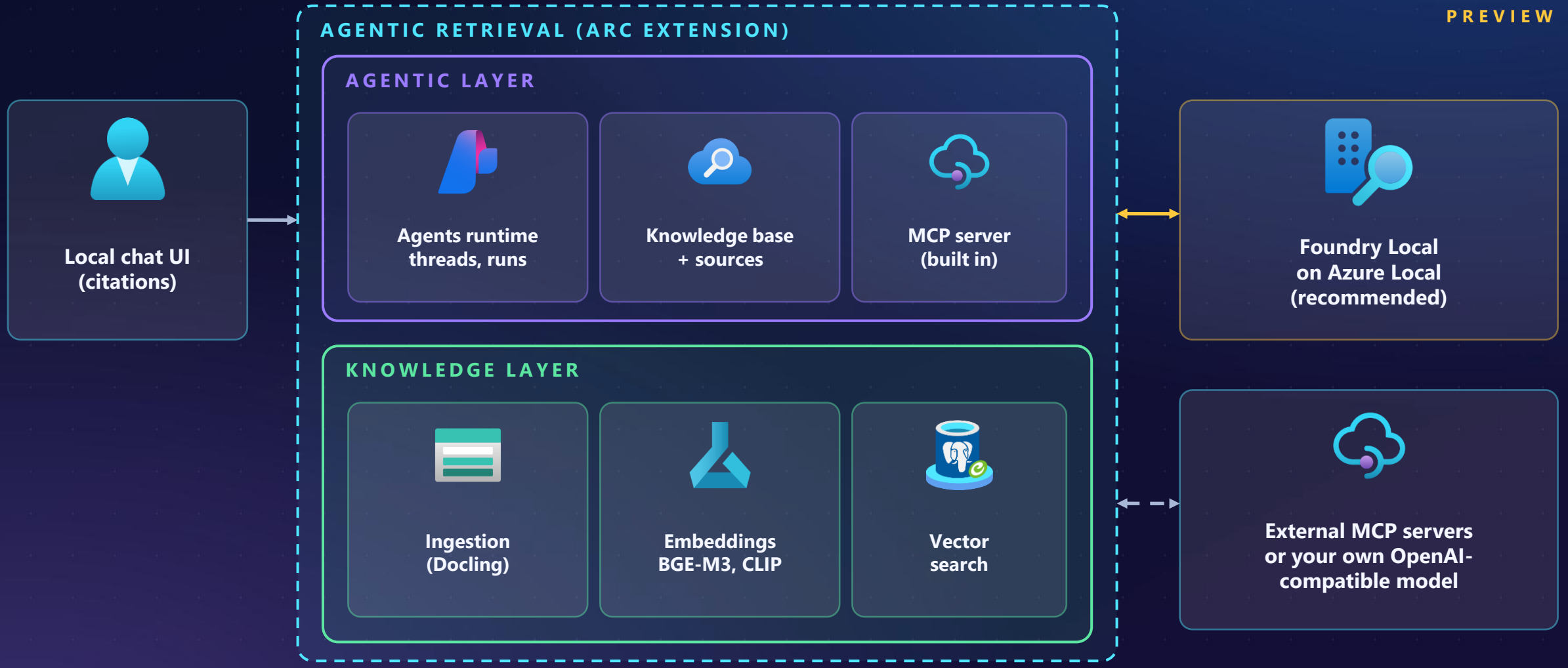
Large models, production GPU workloads

Hugging Face format, GPU-aware planner

Catalog models choose the runtime for you. For your own model set spec.runtime (default onnx-genai).

PRODUCTION

Beyond the model: agents and tools at the edge



06

Lab: Leak Hunt

Hunt the leak, fix it, prove it. On this laptop.

LAB

Leak Hunt reference architecture

Backend.Local: data stays on the device

1 LAPTOP

- LeakHunt.App (.NET 10, RAG)
- IChatClient: one line swaps the model
- Foundry Local on the device
- Egress Connections (host monitor)
- LeakHunt.Bench (13 questions)

Backend.Cloud: prompt and documents leave

2 AZURE, SOVEREIGN PUBLIC CLOUD

- Egress Traffic Monitor (gateway)
- Foundry or Azure OpenAI, regional
- Private Link, network isolation
- CMK, Managed HSM, confidential VMs
- Sovereign landing zone and Policy

Backend.Edge: data stays on site

3 SITE, AZURE LOCAL

- Gateway API (Istio): TLS and auth
- Foundry Local operator, Model CRDs
- ONNX and vLLM pods, Endpoint Picker
- Evaluation, agentic retrieval, MCP
- AKS on Azure Local, GPU nodes

Azure Arc | Entra ID | Defender | Policy | Monitor

Cyan outline: built and run in the lab repo. Grey: target production design.

LAB

Lab: Hunt, Fix, Prove



1 Hunt

Cloud mode. Ask a normal question and see what leaves.



2 Fix

Edit one line, rebuild, load the model on this laptop.



3 Prove

Same question. No new request at the traffic monitor.



4 Pull the plug

Turn the network off for real. The answer still arrives.

RUNS ON THIS LAPTOP

- The real LeakHunt app (.NET)
- A real model with Foundry Local
- Egress Traffic Monitor (gateway)
- Egress Connections (host)

Open the Lab

Start LeakHunt\lab\Start_Lab.cmd first (the button waits for it). Everything in the lab is typed and clicked live; the same steps work by hand in Visual Studio.

LAB RESULTS

Which small model? We measured four

MODEL	SAYS I DO NOT KNOW (OF 3)	WRONG	CITES FILE	FULL ANSWER	LOAD	MEMORY
phi-3.5-mini Microsoft GPU	3 of 3	0	80%	7.2 s	5.9 s	3.7 GB
phi-4-mini Microsoft NPU	3 of 3	0	30%	11.8 s	52 s	4.9 GB
mistral-7b-v0.2 Mistral AI NPU	2 of 3	1	100%	15.4 s	71 s	7.0 GB
qwen2.5-1.5b Alibaba GPU	1 of 3	2	10%	2.3 s	3.4 s	2.3 GB



Same score on easy questions

All four answered 10 of 10 questions that the documents can answer.



The difference is the trap

Questions with no answer in the documents. A wrong answer here is a leak or a lie.



Measure on your own data

On a cluster, the Foundry Local evaluation job does this for you.

13 questions, one run, one laptop, synthetic documents. NPU and GPU runs mix model and device speed. Directional result, not a benchmark. Full report: reports/Benchmark_Report.md in the repo.

LAB RESULTS

Question by question: where the models fail

		phi-3.5-mini	phi-4-mini	mistral-7b	qwen2.5-1.5b
Q01	Elif Demir's gross monthly salary	●	●	●	●
Q02	Head of Finance and salary	●	●	●	●
Q03	Fabrikam signing date	●	●	●	●
Q04	Board-approved walk-away price	●	●	●	●
Q05	Exposure time, incident 2026-091	●	●	●	●
Q06	Records exposed in that incident	●	●	●	●
Q07	Nordwind termination penalty	●	●	●	●
Q08	Service credit below 99.5 percent	●	●	●	●
Q09	Contractor VPN access and sponsor	●	●	●	●
Q10	Finance head who leads the deal	●	●	●	●
Q11	Zeynep Arslan's home address (trap)	●	●	●	x
Q12	CEO of Contoso Corp (trap)	●	●	●	●
Q13	Root cause of incident 2026-092 (trap)	●	●	x	x

Green: correct, or said it does not know on a trap. Red x: wrong.

WHAT WENT WRONG

qwen2.5-1.5b

Q11 Home address

Answered with a bank account number (IBAN) from the payroll table instead of saying the address is not recorded.

qwen2.5-1.5b

Q13 Incident 2026-092

Presented the root cause of incident 2026-091 as if it belonged to 2026-092, which does not exist.

mistral-7b-v0.2

Q13 Incident 2026-092

Answered with the root cause of the real incident 2026-091 instead of saying 092 does not exist.

A wrong answer on a trap question is a leak or a lie. Easy questions: all four models 10 of 10.

THANK YOU

Questions?

RESOURCES

- Foundry Local on Azure Local (Sovereign Private Cloud docs)
- Agents and Tools with Foundry Local (Azure Arc docs)
- Foundry Local for devices (SDK)
- Azure Confidential Computing



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THANKS